

0 8 . 1

The genetic diversity of species is measured by comparing differences in the base sequence of DNA or differences in the base sequence of mRNA.

Give **two** other ways in which genetic diversity between species is measured.

[2 marks]

1

2

Scientists investigated differences between 260 North American bird species by comparing the base sequence of a gene in mitochondrial DNA. They compared the gene base sequence of each bird with all of the other 259 species. For each comparison they calculated the percentage difference in base sequence.

0 8 . 2

Figure 9 shows the base sequence for part of the gene in two species.

Figure 9

Species 1 A G C T G C C T A G A

Species 2 A T G T G G C A A G A

Calculate the percentage difference in base sequence for these base sequences.

[1 mark]

Answer = _____ %

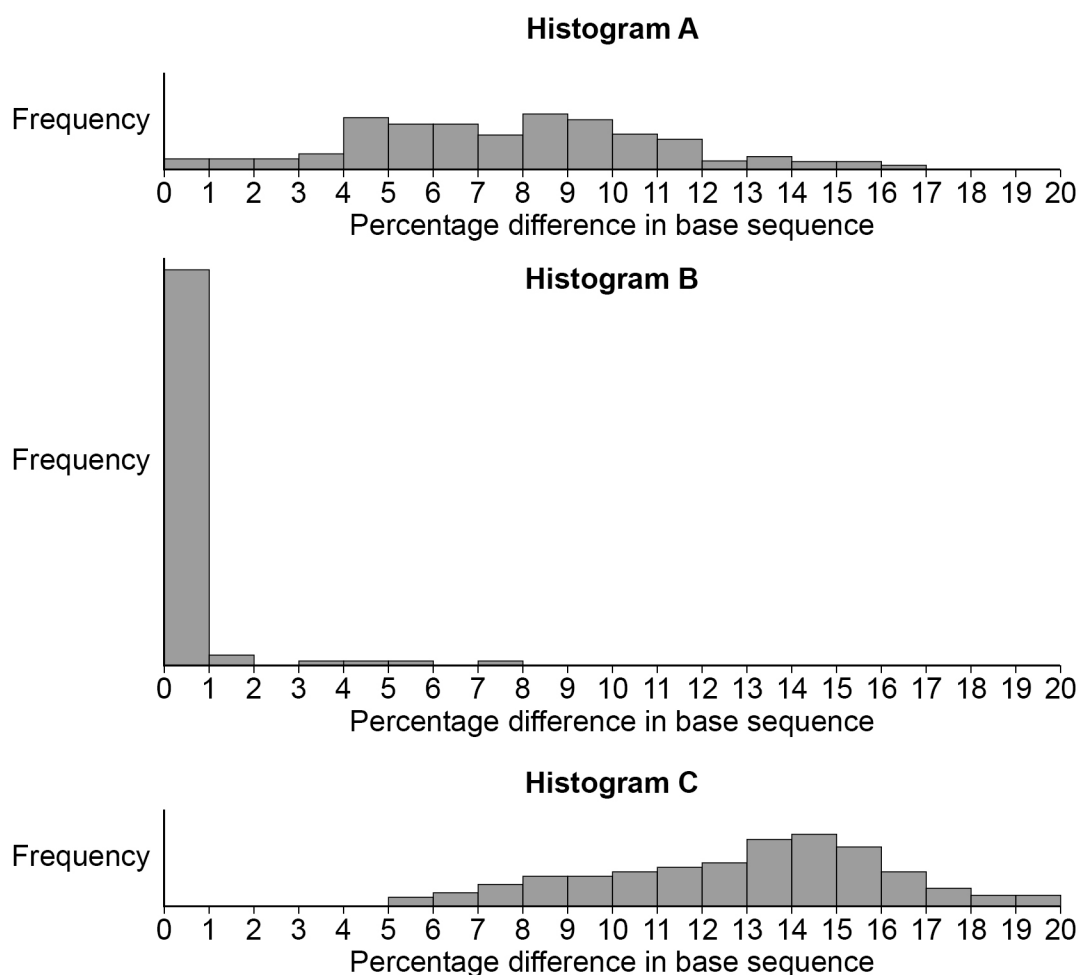


The scientists compared base sequences in:

- birds of the same species
- birds of different species in the same genus
- birds of different species in the same family.

The scientists' results are shown in **Figure 10**.

Figure 10



0 8 . 3

Complete **Table 3** by writing **A, B** or **C** in the box to correctly match the statement to each histogram shown in **Figure 10**.

[1 mark]

Table 3

Statement	Histogram
Base sequences of birds of the same species.	
Base sequences of birds of the same genus.	
Base sequences of birds of the same family.	

Question 8 continues on the next page

Turn over ►



0 8 . 4

To calculate the percentage difference in base sequences, the scientists first counted the number of bases and the number of base differences.

What statistical test should the scientists use to test whether the number of base differences between birds in histogram **A** (on page 25) and birds in histogram **C** (on page 25) is statistically significant?

Place a tick (✓) in the box against the statistical test you would use.

Justify your answer.

[2 marks]

Chi-squared

☐

Correlation coefficient

☐

Student's t-test

☐

Justification

6

